Technical Appendix for
Can Molecular Evolution Mechanism Enhance Molecular Representation?

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A. Methods

A.1. Algorithm Complexity for MEvoN Construction

Denote T as the similarity computing time complexity. The traditional approach requires $O(N^2T)$ similarity computations for N samples. Our two-stage grouping method (Section 2.3) reduces complexity to $O(\frac{N^2\cdot(T_{s1}+(1-\alpha)T_{s2})}{k})$. The first stage employs fast screening (e.g., string matching) to filter out $\alpha \in (0,1)$ molecular pairs ($\alpha \approx 90\%$ for QM9), while the second stage employs more complex graph-based screening to handle complex molecular comparisons. making computational requirements significantly lower than mainstream pre-training methods.

A.2. MEvoN Construction Algorithm

To construct the Molecular Evolution Network (MEvoN), we design a hierarchical similarity-based algorithm that establishes directed links between molecules differing in atomic count. Given a molecular set $\mathcal{M}$ and similarity thresholds θ_1 and θ_2, the algorithm outputs a network $\mathcal{N} = (\mathcal{V}, \mathcal{E})$ comprising molecules as nodes and evolutionarily plausible relations as edges. The algorithm proceeds as follows:

- Molecules are first grouped according to their atom counts, such that $G_k = \{m_i \in \mathcal{M} \mid \mathrm{N}_{\text{atoms}}(m_i) = k\}$.
- For each pair of groups (G_i, G_j) with $i < j$, pairwise molecular similarity is computed using edit-distance or fingerprint-based metrics.
- Pairs exceeding threshold θ_1 are retained in an initial candidate set Pair^1.
- From Pair^1, the subset of pairs with the highest similarity values, denoted as $\text{Pair}^1_{\text{max}}$, undergoes a secondary screening based on Weisfeiler-Lehman graph kernel similarity.
- Pairs surpassing the second threshold θ_2 are recorded in Pair^2 and incorporated into the evolving network by updating $\mathcal{V}$ and $\mathcal{E}$.

This two-stage similarity filtering ensures that only structurally and topologically consistent pairs are connected in MEvoN, yielding a reliable and evolutionarily interpretable molecular graph.

Algorithm 1: MEvoN Construction Algorithm

Input: Set of molecules $\mathcal{M}$.
Parameter: Similarity thresholds θ_1 and θ_2.
Output: A MEvoN $\mathcal{N} = (\mathcal{V}, \mathcal{E})$.

Group molecules by atomic count:
    $G_k = \{m_i \mid \mathrm{N}_{\text{atoms}}(m_i) = k, m_i \in \mathcal{M}\}$
for each pair of evolutionary groups (G_i, G_j), $i < j$ **do**
  Initialize: $\text{Pair}^1 = \emptyset$, $\text{Pair}^2 = \emptyset$
  for each molecule pair $(m_p \in G_i, m_q \in G_j)$ **do**
    Calculate the similarity $S_{\text{edit/fp}}(m_p, m_q)$
    if $S_{\text{edit/fp}}(m_p, m_q) \geq \theta_1$ **then**
      Add (m_p, m_q) to Pair^1
    end if
  end for
  Let $\text{Pair}^1_{\text{max}}$ be the maximum similarity pairs in Pair^1
  if $\left\|\text{Pair}^1_{\text{max}}\right\| > 1$ **then**
    for each pair $(m'_p, m'_q) \in \text{Pair}^1_{\text{max}}$ **do**
      Calculate the similarity $\mathcal{S}_{\text{wl}}(m'_p, m'_q)$
      if $\mathcal{S}_{\text{wl}}(m'_p, m'_q) \geq \theta_2$ **then**
        Add (m'_p, m'_q) to Pair^2
      end if
    end for
  end if
  $\mathcal{V}' \leftarrow \left\{m'_p, m'_q \mid (m'_p, m'_q) \in \text{Pair}^2\right\}$
  $\mathcal{E}' \leftarrow \left\{(m'_p, m'_q) \mid (m'_p, m'_q) \in \text{Pair}^2\right\}$
  $\mathcal{N} \leftarrow (\mathcal{V} \cup \mathcal{V}', \mathcal{E} \cup \mathcal{E}')$
end for

A.3. MEvoN-MPP's Framework

Figure 1 illustrates the architecture of **MEvoN-MPP**, which is composed of two key modules: the path-aware module **EvoP** and the label-aware module **EvoA**.

The **EvoP module** captures the evolutionary relationships between molecular structures. It receives path features with shape $(B, 9, 512)$, embeds them into a lower-

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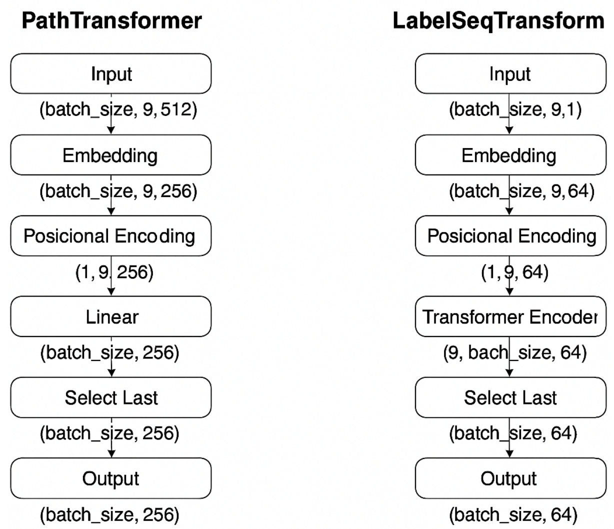


Figure 1: Architectural overview of MEvoN-MPP, highlighting the **EvoP** (Path-Aware) and **EvoA** (Label-Aware) modules. Both components use Transformer-based encoders to process molecular evolutionary information.

dimensional latent space, and processes them using a multi-layer Transformer encoder. A learnable positional encoding is added to retain sequence order. The final output, a 256-dimensional vector, summarizes the evolutionary trajectory for each molecule.

The **EvoA module** incorporates label supervision to inform the weighting of evolutionary paths. Specifically, scalar label sequences of shape $(B, 9, 1)$, such as intermediate energies or properties, are linearly embedded to a 64-dimensional space and passed through a Transformer encoder with attention masking. The last token's representation is used to generate a label-aware evolutionary representation.

Together, EvoP and EvoA enable the model to leverage both structural and supervised signals during molecular evolution, enhancing the learning of property-sensitive features. The outputs from both modules can be fused or further processed with the **MolE module**, which encodes graph-level molecular structures. MolE is flexible and can be instantiated as any molecular encoder, such as GNNs, CNNs, or Transformer-based models.

Table 1 summarizes the configurations of the **EvoP** and **EvoA** modules. Both modules rely on Transformer encoders with 2 layers and 4 attention heads. EvoP focuses on structural path evolution, while EvoA integrates label information to enrich the representation of molecular change.

This evolution-aware representation learning design allows MEvoN-MPP to model the complex progression of molecular structures and properties, providing a principled and scalable approach for molecular property prediction.

B. Results

B.1. Baselines

The experimental comparison methods of QM7 and QM9 in this paper include mainstream and advanced methods from 2017 to 2024. In recent years, the main research directions of methods mainly focus on data pre-training, additional knowledge supplement, plug-in model, and so on.

As a plugin model, our comparison experiments demonstrate two key advantages:

Against SOTA pre-training methods (GEM, Uni-Mol, etc.): Our plugin achieves superior predictive performance *without requiring pre-training*.

- GROVER (Rong et al. 2020) is a self-supervised Transformer trained on millions of unlabeled molecules. It combines local message passing with global attention and employs multiple pretraining objectives across node, edge, and graph levels. GROVER enhances molecular representation learning and consistently improves downstream property prediction tasks when fine-tuned.
- GEM (Fang et al. 2022) augments molecular graphs with auxiliary geometric information by constructing atom–bond and bond–angle graphs. It introduces geometric self-supervised tasks, such as predicting pairwise distances and angles, to guide representation learning. GEM effectively leverages 3D structural information to boost performance across multiple property prediction benchmarks.
- MolCLR (Wang et al. 2022b) is a contrastive learning framework for pretraining graph neural networks on molecular data. It generates augmented molecular views using chemically plausible transformations (e.g., atom masking, subgraph removal) and trains the encoder to align representations of augmented pairs. MolCLR significantly improves GNN generalization on various property prediction tasks after fine-tuning.
- Uni-Mol (Zhou et al. 2023) is a universal 3D molecular representation learning framework built on SE(3)-equivariant Transformer architectures. Pretrained on large-scale molecular conformer and protein pocket datasets, Uni-Mol supports tasks including molecular property prediction, 3D generation, and drug–target interaction modeling. It achieves SOTA results across MoleculeNet (Wu et al. 2018) and molecular docking benchmarks.

Against SOTA backbone methods (ComENet, SchNet, VisNet, etc.): Our plugin enhances performance when integrated *with established mainstream backbones*.

- SchNet (Schütt et al. 2018) is a continuous-filter convolutional neural network designed for quantum molecular property prediction. It directly incorporates interatomic distances via continuous filters, enabling rotational and translational invariance.
- ComENet (Wang et al. 2022a) improves upon conventional message passing by explicitly incorporating spatial geometric relations along with topological graph structure.

Component	EvoP (Path-Aware)	EvoA (Label-Aware)
Input Shape	$(B, 9, 512)$	$(B, 9, 1)$
Embedding Dim	256	64
Positional Encoding	Learnable, $(1, 9, 256)$	Learnable, $(1, 9, 64)$
# Transformer Layers	2	2
# Attention Heads	4	4
Output Dim	256	64

Table 1: Module configuration in MEvoN-MPP.

- Equiformer (Liao and Smidt 2023) is an SE(3)-equivariant graph Transformer that uses tensor field networks to preserve equivariance under spatial transformations. It encodes atomic features as irreducible representations and generalizes attention mechanisms to operate over these tensors.
- ViSNet (Wang et al. 2024) introduces a vector-scalar interactive message passing mechanism that separates and jointly updates scalar features (e.g., atomic types) and vector features (e.g., directions). Its SE(3)-equivariant architecture ensures physically consistent representation learning, while efficiently capturing geometric details such as bond angles and torsions.

B.2. Computing Infrastructure

All experiments were conducted on a high-performance computing server with the following specifications:

- **CPU:** Dual-socket AMD EPYC 7763 64-Core Processor, totaling 256 logical processors (128 physical cores, 2 threads per core), base frequency 1.5 GHz, boost up to 3.53 GHz.
- **GPU:** NVIDIA RTX A6000.
- **Memory:** 256 GB RAM.
- **Cache:** 512 MB L3, 64 MB L2, and 8 MB L1 (aggregated).
- **Operating System:** Ubuntu 20.04 LTS.
- **Software Libraries:** Python 3.9, PyTorch 2.1.0, DGL 1.1.2, RDKit 2023.03.1, NumPy 1.24.3, SciPy 1.10.1, scikit-learn 1.2.2.

B.3. Comparison on Loss Curves of Four Baseline Models

We present the loss curves of four baseline models (i.e., models specifically designed for molecular representation without pre-training) on the test dataset during the training phase, visualized using $\varepsilon_{\text{HOMO}}$, $\varepsilon_{\text{LUMO}}$, and $\Delta\varepsilon$ as shown in Figure 2, Figure 3, Figure 4, and Figure 5. These loss curves illustrate the models' ability to fit different molecular properties and their convergence speed during training. The batch size is set to 1024, and the learning rate is set to 0.0005. We observe that MEvoN-MPP shows a distinct advantage in terms of generalization performance when applied to GCN (Kipf and Welling 2017) and GIN (Xu et al. 2019) models. Compared to the original SchNet (Schütt et al. 2018) and ComENet (Wang et al. 2022a) models, MEvoN-MPP excels in both training effectiveness and convergence. Specifically, when used as a plugin, our method significantly accelerates the convergence speed at the same number of epochs.

B.4. Analysis of Property Variation under Structural Changes (Case Study: Atom Addition)

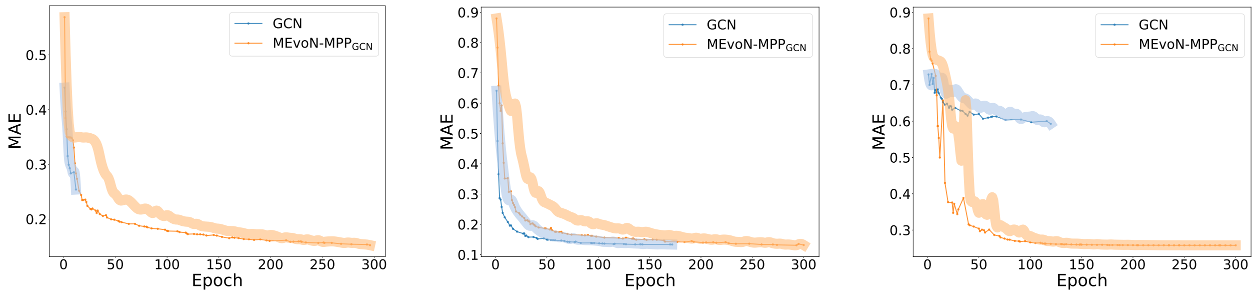
In this section, we analyze how molecular property changes correspond to structural modifications, with a specific focus on atom addition operations. We choose the HOMO-LUMO gap (GAP) as the target property from the QM9 dataset, which contains 12 quantum chemical properties. GAP is known to be particularly sensitive to structural and electronic configuration changes, making it a suitable candidate for this analysis.

We adopt basic GCNs as the prediction model, trained on molecular graphs constructed from SMILES representations. The test set consists of 10% of the entire QM9 dataset, randomly selected to ensure sufficient diversity and coverage. The objective is to examine how well the model captures property variations induced by controlled modifications in molecular structure.

We construct evolutionary paths between molecular pairs where the only modification is the addition of atoms. In total, we extract 17 distinct atom addition combinations, such as adding a single carbon atom ("Added atoms: C:1"), adding oxygen and nitrogen together ("Added atoms: O:1, N:1"), and so on. Using RDKit, we parse SMILES strings to compute atomic composition differences and categorize each molecular pair by its structure change type.

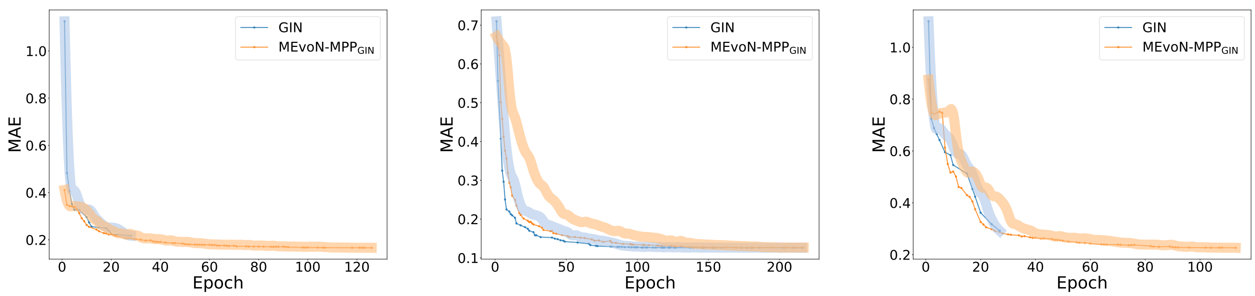
For each such molecular transformation, we compute both the true GAP change and the GCN-predicted GAP change. By comparing these values across the 17 atom addition types, we assess the model's ability to generalize under systematic structure variations. We observe that GCN performs reasonably well for simpler changes (e.g., adding a single carbon atom), where the electronic effect is relatively localized. However, for more complex changes involving multiple heteroatoms, the model's predictions deviate more significantly from the ground truth, suggesting limitations in representing the induced structural–electronic coupling using only 2D graph topology.

Figure 6 to 22 presents the distribution of GAP changes under the 17 atom addition scenarios. Each subplot corresponds to a distinct structure change type, with the x-axis denoting molecular pairs and the y-axis indicating the GAP value change. Ground truth and predicted values are visualized using different colors for direct comparison. From



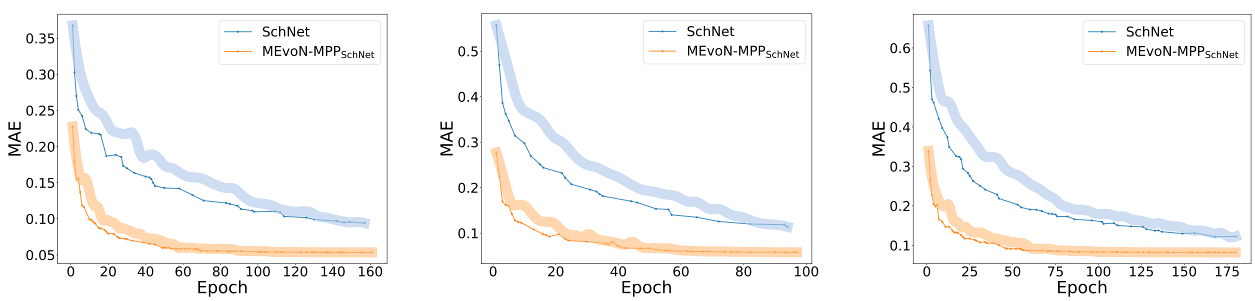
(a) MAE for $\varepsilon_{\mathrm{HOMO}}$ (b) MAE for $\varepsilon_{\mathrm{LUMO}}$ (c) MAE for $\Delta\varepsilon$

Figure 2: MEvoN$_{\mathrm{GCN}}$'s results on the MAE for $\varepsilon_{\mathrm{HOMO}}$, $\varepsilon_{\mathrm{LUMO}}$, and $\Delta\varepsilon$.



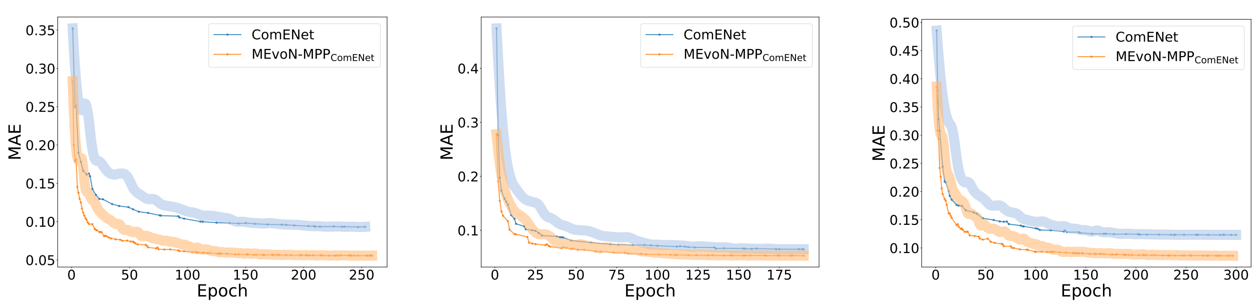
(a) MAE for $\varepsilon_{\mathrm{HOMO}}$ (b) MAE for $\varepsilon_{\mathrm{LUMO}}$ (c) MAE for $\Delta\varepsilon$

Figure 3: MEvoN$_{\mathrm{GIN}}$'s results on the MAE for $\varepsilon_{\mathrm{HOMO}}$, $\varepsilon_{\mathrm{LUMO}}$, and $\Delta\varepsilon$.



(a) MAE for $\varepsilon_{\mathrm{HOMO}}$ (b) MAE for $\varepsilon_{\mathrm{LUMO}}$ (c) MAE for $\Delta\varepsilon$

Figure 4: MEvoN$_{\mathrm{SchNet}}$'s results on the MAE for $\varepsilon_{\mathrm{HOMO}}$, $\varepsilon_{\mathrm{LUMO}}$, and $\Delta\varepsilon$.



(a) MAE for $\varepsilon_{\mathrm{HOMO}}$ (b) MAE for $\varepsilon_{\mathrm{LUMO}}$ (c) MAE for $\Delta\varepsilon$

Figure 5: MEvoN$_{\mathrm{ComENet}}$'s results on the MAE for $\varepsilon_{\mathrm{HOMO}}$, $\varepsilon_{\mathrm{LUMO}}$, and $\Delta\varepsilon$.

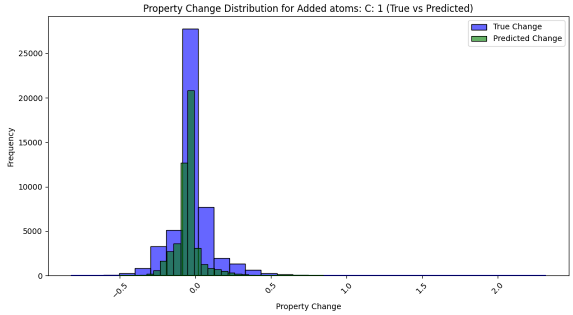


Figure 6: Property changes for structure change type: Added atoms C:1. True values vs. model predictions.

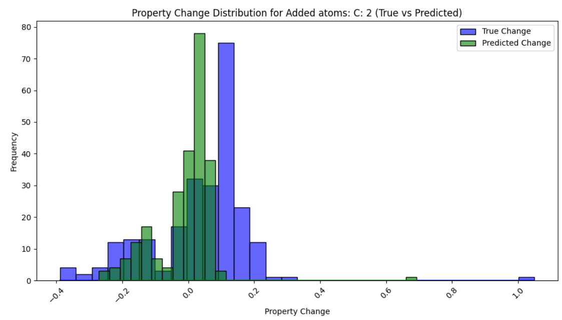


Figure 7: Property changes for structure change type: Added atoms C:1. True values vs. model predictions.

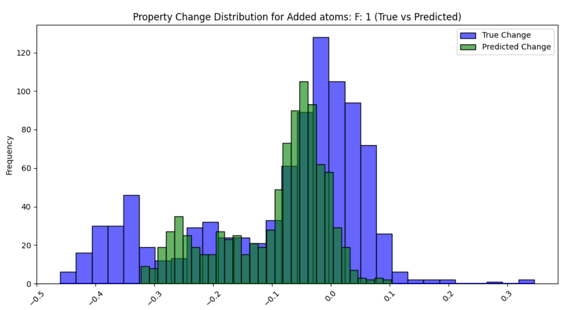


Figure 8: Property changes for structure change type: Added atoms C:1. True values vs. model predictions.

the plots, it is evident that structural changes affect the target property in highly variable ways, and GCN's ability to track these effects depends strongly on the complexity of the atomic addition pattern.

This analysis highlights that while GCN can effectively model simple atom-level modifications, it struggles with more nuanced or multi-atom structural changes. Future work may involve incorporating richer 3D geometric information or employing equivariant architectures (e.g., Equiformer, SchNet, or ViSNet) to improve prediction robustness under such chemically meaningful perturbations.

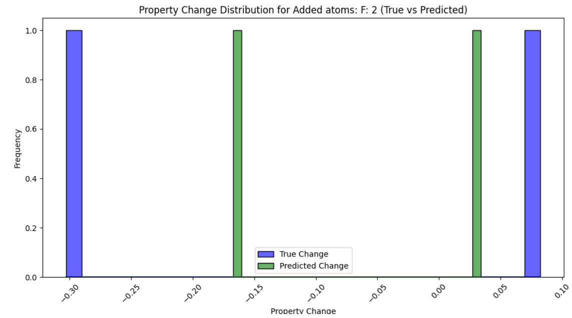


Figure 9: Property changes for structure change type: Added atoms C:1. True values vs. model predictions.

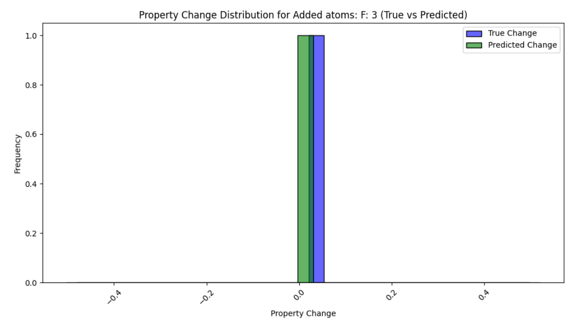


Figure 10: Property changes for structure change type: Added atoms C:1. True values vs. model predictions.

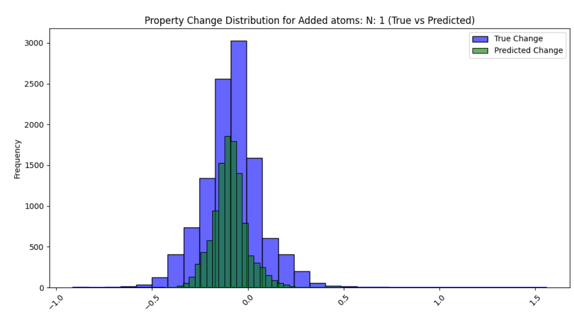


Figure 11: Property changes for structure change type: Added atoms C:1. True values vs. model predictions.

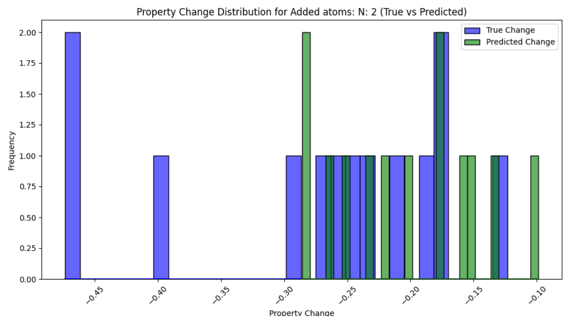


Figure 12: Property changes for structure change type: Added atoms C:1. True values vs. model predictions.

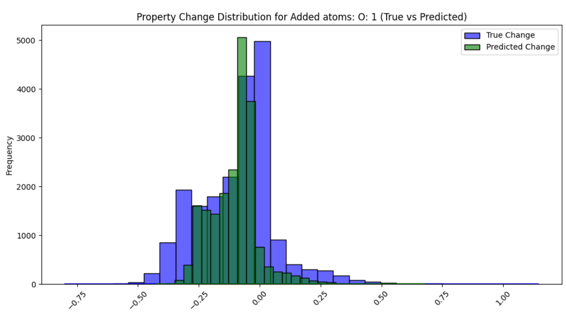


Figure 13: Property changes for structure change type: Added atoms C:1. True values vs. model predictions.

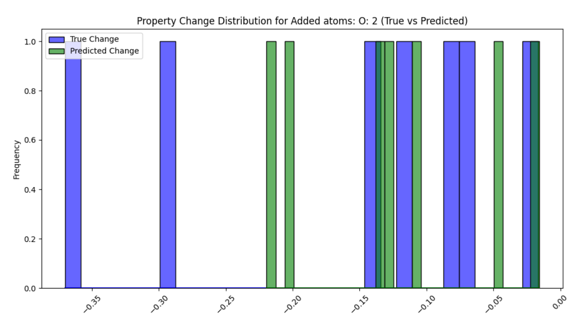


Figure 14: Property changes for structure change type: Added atoms C:1. True values vs. model predictions.

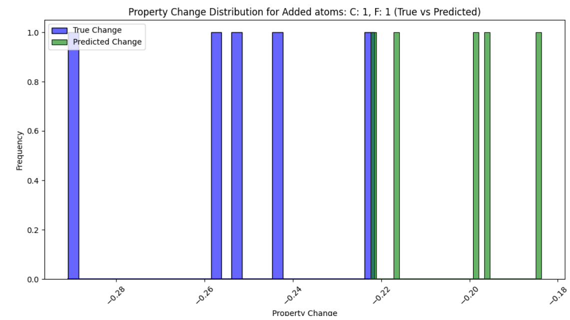


Figure 15: Property changes for structure change type: Added atoms C:1. True values vs. model predictions.

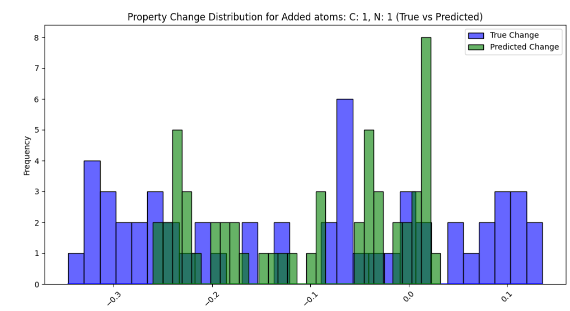


Figure 16: Property changes for structure change type: Added atoms C:1. True values vs. model predictions.

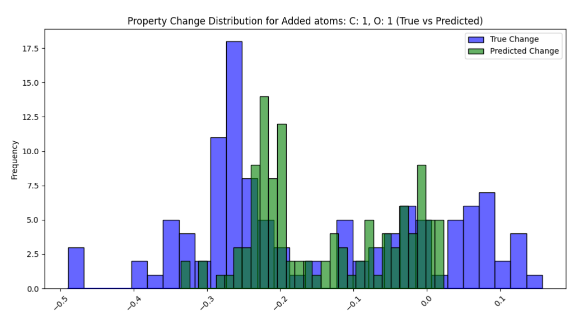


Figure 17: Property changes for structure change type: Added atoms C:1. True values vs. model predictions.

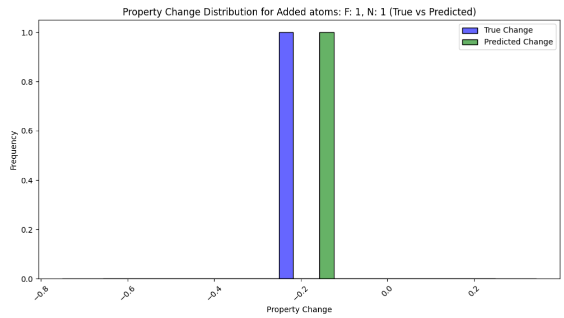


Figure 18: Property changes for structure change type: Added atoms C:1. True values vs. model predictions.

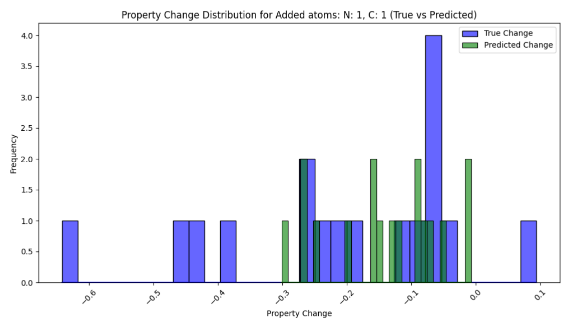


Figure 19: Property changes for structure change type: Added atoms C:1. True values vs. model predictions.

Figure 20: Property changes for structure change type: Added atoms C:1. True values vs. model predictions.

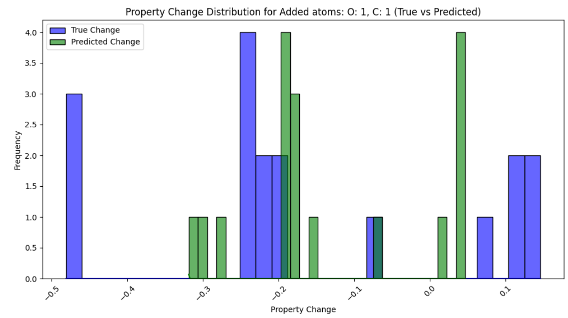


Figure 21: Property changes for structure change type: Added atoms C:1. True values vs. model predictions.

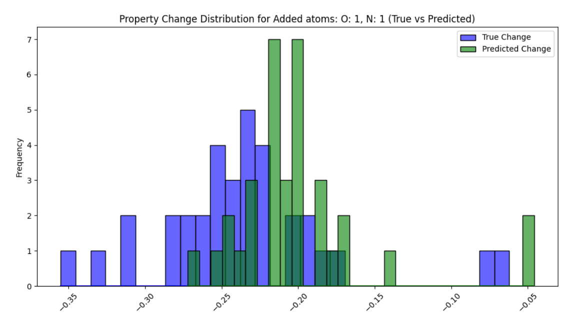


Figure 22: Property changes for structure change type: Added atoms C:1. True values vs. model predictions.

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